COMP0118: Computational Modelling for Biomedical Imaging

Coursework 1

Giuliano Costa 24225233

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Note: I used Plotly thoughout my Jupyter notebook, an interactive plotting library, they may not render correctly depending on your IDE. Also, all code in the notebook is organized by question number, for easy referencing.

1.1 Parameter Estimation

1.1.1 Linear diffusion tensor estimation

Our DWI (Diffusion Weighted Imaging) data consists of 108 discrete measurements of the brain. Each measurement forming a full brain volume with a different measured diffusion direction, but from the same patient and MRI session.

We can map mean diffusivity and fractional anisotropy to visualize the brain structures in slice 72. To do this, we use the linear diffusion tensor estimation. I used NumPy's linear least squares method to solve the linear system and build the diffusion tensor. I then plotted the three maps. (Fig. 1) Our diffusion tensor is successful in revealing the different structures in this plane of the brain.

1.1.2 Ball and Stick Model Fitting

Fitting the ball and stick model the DWI data was performed using non-linear optimization on a single voxel. The model parameter estimates on voxel [:, 91, 64, 71] were $S_0 = 3.52 \times 10^3$, $d = -4.98 \times 10^{-6}$, $f = 1.20 \times 10^2$, $\theta = 8.87 \times 10^{-1}$, and $\phi = 1.56$. These results reveal significant issues with the unconstrained optimization approach. Our RESNORM of 5.56×10^7 is much higher than expected given the noise level ($\sigma \approx 200$). We can calculate our expected RESNORM with $E[RESNORM] = N \cdot \sigma^2 = N \cdot 40,000 = 4.32 \times 10^6$

Our RESNORM is 12.66 times higher than expected from noise alone. This is likely due to the fact that our parameters have unrealistic start values and fitted values. Notably, d is initalized negative and our fitted f is greater than 1 (volume fraction cannot be greater than 1). Our model fails to capture the baseline $b = 0$ signal strengths (see Fig. 2).

1.1.3 Model constraints

To improve our ball and stick model we need to do two things: adjust the start values to be realistic and add transformations to prevent non-physical values during fitting iterations. We can constrain our S_0 and d parameters to be positive by squaring them, and transform f with a sigmoid operation to enforce 0-1 bounds.

We initialize our parameters with the inverse of our transformations and give them physically realistic values. Now d will start positive and stay positive, f will start at 0.5 and stay in a 0-1 range, etc...

```
startx_transformed = np.array([np.sqrt(3500),np.sqrt(5e-6),0,0,0])
```

After these changes, our RESNORM decreased to 5.87×10^6, our d is now positive, and our f is physically accuate:

```
S0: 4.2579e+03 , diff: 1.1413e-03, f: 3.5731e-01, theta: -9.8107e-01,
phi: 5.7945e-01 , RESNORM: 5.8720e+06
```

Plotting our measured and predicted signal values indicates much better fitting of the baseline and diffusion weighted signals. (see Fig. 3)

1.1.4 Global minimum assessment

To determine if our model parameters have converged to a global minimum, we can use NumPy's Gaussian generator to create distributions with means around our existing start values and standard deviation as 10% of the original value. Then we can repeat the fitting process many times with these randomly selected start parameters. Finally, we can calculate the probability of finding our theorized global minimum using the following equation. Where p is the frequency of getting the same minimum:

$$n \geq \frac{\ln(1 - \text{confidence})}{\ln(1 - p)}$$

For our original voxel choice, we find the global minimum is 5.87×10^6 79% of the time in our 100 trials. Meaning we need 2 trials to have a 95% confidence. There is indeed a global minimum for all 4 voxels selected. Suggesting our ball and stick model is consistent. (See Fig. 4)

Voxel	Min RESNORM	Mean RESNORM	"Success" Rate	Trials for 95%
[:,91,64,71]	5.87×10^6	7.76×10^6	79.0%	2
[:,93,63,71]	5.69×10^6	9.51×10^6	64.0%	3
[:,89,34,71]	7.96×10^6	9.00×10^6	46.0%	5
[:,60,65,71]	4.61×10^6	5.80×10^6	82.0%	2

Table 1: Comparison of Ball-and-Stick model fitting performance across different voxels. Each voxel was tested with 100 trials using random starting points. Success rate indicates the percentage of trials that converged to within 0.1% of the minimum RESNORM value.

1.1.5 Parameter mapping

Using a similar method from 1.1.4, the 25,230 voxels in slice 72 were each fitted 5 times (5 trials for a 95% confidence interval was the maximum in our 4 voxel tests) for a total of 126,150 fittings. The best RESNORM was selected and maps were generated for the set of voxels (Fig. 5). The minimization algorithm selected severely impacted performance, L-BFGS-B, while faster and less memory intensive, led to horrendous numerical instability. The derivative-free Nelder-Mead, though slower, led to much more sensible estimations. For the fiber direction plot, (Fig. 6) a mask around the skull (using a diffusivity threshold) was used. Then, fiber directions were transformed to Cartesian coordinates creating an array of 2D vectors. These vectors were weighted by the Volume Fraction f map and overlayed on the f map to provide clarity. The plot provides us insight with the fiber directions, notably along the corpus callosum, where two big "chunks" of vectors can be seen moving bidirectionally between the hemispheres.

1.1.7 Rician Noise Model

Two objective functions were created, an accurate Rician NLL and an approximation from Alexander 2009 using an offset of the Rician bias. With a 16 core (32 threads) AMD Ryzen CPU, results were: Baseline Ball and Stick: 1m 02s, Accurate Rician: 15m 37s, Approximate Rician: 1m 30s. The approximate Rician is 50% slower than regular ball and stick, but creates far fewer artifacts (Fig. 7 & 8). The accurate Rician produces no noticeable improvements.

1.2 Uncertainty Estimation

1.2.1 Bootstrap

We can extend our global minimum assessment by identifying the 2-sigma range and 95% CI for the first three parameters of our Ball and Stick constrained model, $S0$, d, and f. Using the bootstrap method, 3 voxels were bootstrapped 1000 times, the original voxel (WM), one in the cortical GM, and one in the corpus callosum (figures

9-11). Volume fraction f proved to be the most uncertain parameter, whereas S_0 is roughly normal, and d is bimodal. Results were fairly consistent for all three voxels.

Parameter	$\mathbf{S_0}$	**d**	**f**
95% Range	$[2.195 \times 10^3, 2.815 \times 10^3]$	$[1.552 \times 10^{-17}, 4.007 \times 10^{-4}]$	$[0.469, 1.000]$
2σ Range	$[2.195 \times 10^3, 2.815 \times 10^3]$	$[1.552 \times 10^{-17}, 4.007 \times 10^{-4}]$	$[0.469, 1.000]$

Table 2: Bootstrap parameter estimates for original voxel [:,91,64,71]

1.2.2 MCMC

To estimate the 2-sigma range and 95% CI for each parameter, we implemented the Metropolis-Hastings algorithm. MCMC parameter estimates are vastly different from Bootstrapping for the same voxel. They also do not align with our findings in 1.1.4 for the best parameter fit. Most notably, S_0, where MCMC suggests values around 2.4-2.6×10^7 with a narrow spread, while bootstrapping yielded much lower values around 2.2-2.8×10^3 with wider relative uncertainty (Table 3). This could be the result of an incorrect implementation of Metropolis-Hastings. (See Fig. 12)

Parameter	$\mathbf{S_0}$	**d**	**f**
95% Range	$[2.387 \times 10^7, 2.612 \times 10^7]$	$[1.010 \times 10^{-3}, 1.015 \times 10^{-3}]$	$[0.670, 0.684]$
2σ Range	$[2.387 \times 10^7, 2.612 \times 10^7]$	$[1.010 \times 10^{-3}, 1.015 \times 10^{-3}]$	$[0.670, 0.684]$

Table 3: MCMC parameter estimates for original voxel [:,91,64,71]

1.3 Model Selection

1.3.1 Adapt to ISBI data

We can adapt our Ball and Stick constrained model to our new dataset which has a richer set of measurements but is limited to 6 voxels (we only work with the first one). The starting range of S_0 and d were changed to reflect normalization. I used the Nelder-Mead algorithm for 100 fitting steps. The frequency of finding the minimum was 94%. The optimization yielded $S_0 = 1.00$, $d = 3.78 \times 10^{-2}$ mm²/s, $f = 0.57$, $\theta = -1.54$ rad, $\phi = -0.08$ rad, and RESNORM 15.106. With a stated noise standard deviation of 0.04, we would expect a total squared residual of approximately $3612 \times (0.04)^2 = 5.78$. Our actual residual norm being 2.6 times higher suggests model inadequacy. (See Fig. 13 for prediction plot)

1.3.2 Model Fitting

Zeppelin Stick had the lowest frequency of finding the global minimum (38%). This could be a result of incorrect starting parameters or just a bad implementation. Though it did achieve the best fit. (See Fig. 14-16 for prediction plots)

Model	RESNORM	Global Min. Frequency
Zeppelin-Stick	10.82	0.38
Zeppelin-Stick-Tortuosity	11.61	0.77
Ball-Stick	15.11	0.94
Diffusion Tensor	18.19	–

Table 4: All models except Diffusion Tensor were fitted 100 times.

1.3.3 Model Comparison

Comparing our 4 models, the results are more or less what we expected, Diffusion Tensor and Ball and Stick have significantly higher RESNORMs (18.19, 15.11, respectively). Surprisingly, the Zeppelin Stick with Tortuosity fits the ISBI data slightly worse (RESNORM: 11.61) than regular Zeppelin Stick (RESNORM: 10.82). Even though Zeppelin Stick has the additional parameter, λ_2, it was overwhelmingly favored by both AIC and BIC criteria with an Akaike weight of 1.0. Suggesting that the additional parameter enables more flexibility during fitting. (See Table 3)

Model	n params	RESNORM	AIC	ΔAIC	Akaike weight	ΔBIC
Zeppelin-Stick	6	10.82	-20977.07	0.00	1.0	0.00
Zeppelin-Stick-Tortuosity	5	11.61	-20724.91	252.16	0.0	245.97
Ball-Stick	5	15.11	-19772.65	1204.42	0.0	1198.23
Diffusion Tensor	2	18.19	-19107.40	1869.67	0.0	1844.90

Table 5: Comparison of different diffusion models using AIC and BIC criteria. Lower AIC and BIC values indicate better model fit, taking into account model complexity.